

AdaBoost vs Gradient Boosting vs XGBoost

Detailed differences | scikit-learn AdaBoostClassifier and GradientBoostingClassifier; XGBoost XGBClassifier

Aspect	AdaBoost	Gradient Boosting	XGBoost
Full name	Adaptive Boosting	Gradient Boosting	Extreme Gradient Boosting
Core idea	Give more weight to misclassified records.	Add a tree that helps reduce the current loss.	Use gradient boosting with regularization and training controls.
Next learner is guided by	Weights assigned to training records.	Gradient of the loss function.	Gradients and second-order information from the loss.
Teaching phrase	Focus on mistakes.	Reduce the remaining loss.	Reduce loss with more controls.
Typical base trees	Often small trees, including one-split decision stumps.	Usually shallow decision trees.	Usually shallow decision trees.
How trees combine	Weighted votes for classification.	Each tree adds a correction to the model score.	Each tree adds a correction to the model score.
n_estimators	Number of weak learners.	Number of boosting stages.	Number of boosting rounds.
learning_rate	Controls each learner's contribution.	Controls each tree's contribution.	Controls each tree's contribution.
Tree depth	Set on a supplied base decision tree.	max_depth controls the individual trees.	max_depth controls the individual trees.
Loss flexibility	More limited in standard scikit-learn AdaBoostClassifier.	Optimizes a specified classification loss.	Offers objectives for binary and multiclass classification.
Regularization	Fewer built-in controls in the AdaBoost estimator.	Control complexity with tree size, learning rate and related settings.	Explicit L1 and L2 regularization settings.
Row sampling	No subsample setting in standard AdaBoostClassifier.	Has a subsample setting.	Has a subsample setting.
Column sampling	No comparable built-in setting in the standard estimator.	No comparable built-in setting in the standard estimator.	Offers settings such as colsample_bytree.
Missing feature values	Handled in the notebook preprocessing pipeline.	Handled in the notebook preprocessing pipeline.	Can handle missing feature values internally; the notebook uses shared preprocessing.
Categorical text	Encoded in the notebook.	Encoded in the notebook.	One hot encoded in the notebook; other categorical support requires configuration.
Main tuning task	Balance learner count, learning rate and base tree complexity.	Balance depth, trees, learning rate and sampling.	More controls offer flexibility but enlarge the tuning task.
Training style	Sequential boosting.	Sequential boosting.	Efficient, optimized tree-building implementation.
Good first lesson	Why later learners focus on difficult cases.	How a loss decreases step by step.	How tuning controls affect an applied model.
Possible limitation	May focus strongly on difficult or mislabeled records.	Can overfit with complex trees or too many rounds.	Can also overfit with poorly chosen settings.
Best on every dataset?	No.	No.	No.

Gradient Boosting vs XGBoost

Focused comparison | scikit-learn GradientBoostingClassifier and XGBoost XGBClassifier

Aspect	Gradient Boosting	XGBoost
Relationship	General gradient boosting approach implemented by scikit-learn.	An optimized implementation of gradient boosted decision trees.
How a new tree is fitted	Fits a tree to the negative gradient of the chosen loss.	Uses gradients plus second-order (curvature) information to choose splits and leaf values.
Main goal	Each new tree reduces the remaining prediction error measured by the loss.	Each new tree reduces the loss while controlling model complexity.
Regularization	Controls complexity mainly through tree settings, learning rate and sampling.	Adds explicit L1 and L2 penalties; also controls splits through gamma and minimum child weight.
Column sampling	No built-in column-sampling setting in this estimator.	Can sample features for each tree, level or split.
Row sampling	subsample selects a fraction of training rows for each stage.	subsample selects a fraction of training rows for each round.
Missing numerical values	Typically requires imputation before fitting.	Learns a default split direction for missing values.
Training speed	Often slower on large datasets with the standard estimator.	Optimized tree building and parallel computation often help on larger datasets.
Tuning complexity	Fewer options: easier to introduce and tune.	More controls: greater flexibility and more tuning choices.
Typical Python class	sklearn.ensemble.GradientBoostingClassifier	xgboost.XGBClassifier
When to demonstrate	Show how sequential trees correct current predictions.	Show how regularization, feature sampling and system optimizations extend boosting.

Simple takeaway: XGBoost is a gradient boosting implementation with additional training and tuning controls.